

CSP529 Technical Writing & Publishing W26

Practical 2a – Extension

From LaTeX Elements to a Professional Technical Document

Technical content drives the document; LaTeX gives it structure.

*** Text mentioned in this color (purple) are to be submitted as a part of this practical.**

We can create a table, equation, figure and algorithm in LaTeX—but can we make them work together to communicate one technical idea?

That is substantially more meaningful than simply "learn more LaTeX commands."

Stage 1 — Inspect Previous Work of

Practical 2.

Each group opens the document produced in Practical 2 and examines it critically.

Now, identify:

1. Which LaTeX constructs did we actually use?
2. Which were difficult?
3. Which elements were merely inserted because the assignment required them?
4. Which elements actually helped communicate the technical concept?
5. Are the figure, table, equation and algorithm referred to from the text?

The crucial rule is:

Nothing should appear in a technical document without being introduced, referred to, or interpreted in the text.

For example, not merely:

[Table 1 appears]

but:

"Table 1 compares the computational characteristics of the three cases..."

Stage 2 — Build a Proper Document Structure

Without using a template, every group to restructure its existing document using:

Title → Abstract → Introduction → Technical Background → Analysis/Working → Discussion → Conclusion → References

You must decide what existing material belongs where.

A) Submit a short justification:

Why did you place each type of technical content in the particular section?

Stage 3 — Master Figures and Tables

Now concentrate specifically on floats.

Every group must include:

- one figure;

- one table;
- captions;
- labels;
- cross-references from the text.

Require the text to contain sentences such as:

"The computational steps are illustrated in Figure 1."

and

"The comparison presented in Table 1 indicates..."

B) Each group must discover:

- How is a figure numbered automatically?
- How is a table numbered automatically?
- What happens if another figure is inserted before Figure 1?
- Why should we use `\ref{ }` instead of manually writing "Figure 1"?
- Why are captions important?

Could you get the feel of **why of cross-referencing?**

Stage 4 — Mathematics Is Part of the Argument

Each group must include at least **two meaningful equations**, not equations inserted merely to satisfy the assignment.

They must:

1. number at least one equation;
2. label it;
3. refer to it from the text;
4. interpret it in words.

C) What does the equation communicate that the paragraph alone does not?

Stage 5 — Algorithm vs. Explanation

Requires:

Technical explanation → Algorithm/ Pseudocode → Explanation of important steps

D) If the algorithm is already given, why do we still need explanatory text?

This should generate a useful discussion.

Please record it and submit.

Could you discover that pseudocode describes **procedure**, while prose can communicate **purpose, rationale, assumptions and interpretation**?

Stage 6 — Cross-Referencing Challenge

Require every group's document to contain cross-references to at least:

- 2 sections
- 2 figures/tables
- 1 equation
- 1 algorithm

Insert a new section and one new figure somewhere in the middle of your document. Recompile.

E) Q1. What changed automatically?

Q2. Why professional technical documents use logical references rather than manually typed numbering?

Stage 7 — References: The Mechanics

Require **three genuine references** connected with your technical content.

You should learn:

- `\cite{}`
- bibliography entries
- citation appearing in text
- reference appearing in the reference list
- difference between **citation** and **reference**

Today we are learning HOW LaTeX manages references. Next Monday we will ask the much harder question: WHY should a particular source be trusted and cited?

Stage 8 — LaTeX Debugging Challenge

Each group records **three errors encountered during today's work**:

Error/ problem	What caused it?	How was it corrected?
Figure not appearing	?	?
Undefined reference	?	?

Missing package	?	?
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Use **your actual compilation problems**.

This directly feeds the Writing Logbook already established in Practical 2.

Final Deliverable

A 3–4 page improved version of your Practical 2 document.

It must contain:

- structured sections and subsections;
- meaningful title and abstract;
- at least one figure with caption, label and textual reference;
- at least one table with caption, label and textual reference;
- at least two equations, including one numbered and cross-referenced equation;
- one algorithm/pseudocode;
- itemized/enumerated material where genuinely appropriate;
- section/figure/table/equation/algorithm cross-references;
- at least three citations and references;
- conclusion;
- consistent professional formatting.

No credit simply for adding an element. The element must contribute to communicating the technical concept.

Final Challenge

After completing the document, each group to exchange the **PDF only—not the .tex source**—with another group.

The reviewing group identifies:

F) "Five places where LaTeX helped technical communication"
and

G) "Three places where LaTeX was used correctly but the technical communication was still weak."

Can you say that correct LaTeX does not guarantee good technical writing?

Individual reflection question:

"What can LaTeX do for a technical writer, and what can LaTeX never do for a technical writer?"
